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LLaVA-CoT: Let Vision Language Models Reason Step-by-Step

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Abstract

Large language models have demonstrated substantial advancements in reasoning capabilities. However, current Vision-Language Models (VLMs) often struggle to perform systematic and structured reasoning, especially when handling complex visual question-answering tasks. In this work, we introduce `LLaVA-CoT`[1]*, a large VLM designed to conduct autonomous multistage reasoning. Unlike chain-of-thought prompting,* `LLaVA-CoT` *independently engages in sequential stages of summarization, visual interpretation, logical reasoning, and conclusion generation. This structured approach enables* `LLaVA-CoT` *to achieve marked improvements on reasoning-intensive tasks. To accomplish this, we construct the* `LLaVA-CoT-100k` *dataset, integrating samples from various visual question answering sources and providing structured reasoning annotations. Besides, we propose a test-time stage-wise retracing search method (SWIRES), which enables effective and efficient test-time scaling. Remarkably, with only 100k training samples and test-time scaling,* `LLaVA-CoT` *not only outperforms its base model by* ***9.4%*** *on a wide range of multimodal reasoning benchmarks, but also surpasses the performance of larger and even closed-source models, such as Gemini-1.5-pro, GPT-4o-mini, and Llama-3.2-90B-Vision-Instruct. The code, dataset, and pre-trained weights are publicly available at https://github.com/PKU-YuanGroup/LLaVA-CoT.*

1. Introduction

Large language models, represented by OpenAI o1 [68] and Deepseek R1 [15], demonstrate strong capabilities for systematic and in-depth reasoning, validating the effectiveness of test-time scaling for language models [51]. However, vision is equally important for enabling models to fully understand the world and extend their cognitive abilities [6, 65].

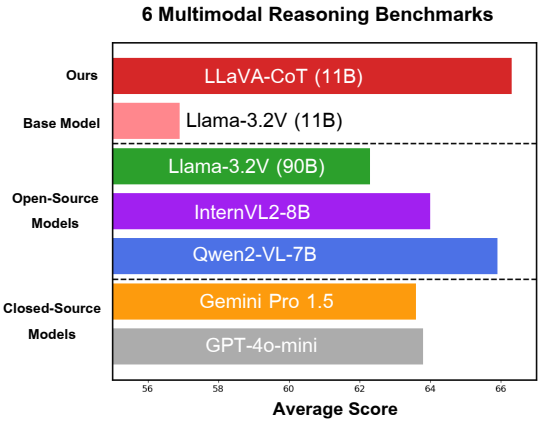


Figure 1. **Performance of LLaVA-CoT and other models across six multimodal reasoning benchmarks.** Although `LLaVA-CoT` is fine-tuned from the Llama-3.2-11B-Vision-Instruct [43] model (which has the lowest average score), it outperforms many larger open-source models and even some closed-source models. Detailed benchmark results are shown in Table 5.

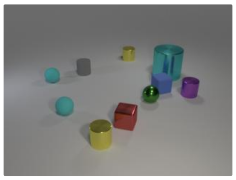
Therefore, developing a multimodal model that integrates both language and vision while facilitating effective, systematic, and deep reasoning holds substantial significance.

Early open-source vision language models (VLMs) mainly employ a direct prediction approach, generating brief answers immediately in response to a question [26, 33, 35]. The main limitation of this direct-response paradigm is its lack of a structured reasoning process, making it less effective for tasks demanding logical reasoning [66]. Recent studies have shown that incorporating Chain-of-Thought (CoT) reasoning encourages the model to reason step by step, significantly improving its question-answering capabilities [56]. However, even with CoT reasoning, most VLMs frequently produce errors or hallucinated outputs during the reasoning progress [29, 34, 53].

Our findings suggest that a significant cause of these issues is the insufficiently systematic and structured nature of

*G. Xu, P. Jin, and Z. Wu contribute equally.
[1]Our `LLaVA-CoT` is built upon Llama-3.2-Vision model [43].

Example 1: Reasoning Problems



Question: Subtract all tiny shiny balls. Subtract all purple objects. How many objects are left?

Options: A. 4, B. 8, C. 2, D. 6

Answer from Llama-3.2-11B-Vision-Instruct:

To determine the correct answer, we need to follow the instructions provided.

The first instruction is to subtract all tiny shiny balls. Since there are **three tiny shiny balls** in the image, we would be left with no shiny balls after performing this action.
The second instruction is to subtract all purple objects. However, upon closer inspection, it becomes apparent that there are **no shiny purple objects present in the image**. Therefore, we cannot subtract any purple objects.

Given that we have already removed all the tiny shiny balls and there were no purple objects to begin with, the correct answer is C. 2, as initially, **there were 2 purple objects, but they are not to be subtracted** according to the instruction.

Answer from LLaVA-CoT:

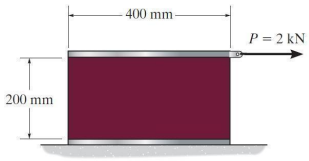
Summary stage: What's the problem? What should I do?
<SUMMARY> I will examine the image to identify and count all the objects, **focusing on the tiny shiny balls and purple objects**. Then, I will subtract these objects from the total count and **determine the number of remaining objects**. *</SUMMARY>*

Caption stage: What can I know from the image?
<CAPTION> The image shows various 3D geometric shapes on a flat surface, including spheres, cubes, and cylinders in different colors. Notably, there are **small shiny green spheres and a purple cylinder**. *</CAPTION>*

Reasoning stage: How to solve the problem step-by-step?
<REASONING> First, I will count all the objects in the image: there are **10 objects in total**.
Next, I will identify and subtract the tiny shiny balls. There is **one small shiny green sphere**.
Then, I will identify and subtract the purple objects. There is **one purple cylinder**.
By subtracting the two objects (the shiny ball and the purple cylinder) from the total, I find that **8 objects remain**. *</REASONING>*

Conclusion stage: What is the final answer?
<CONCLUSION> B *</CONCLUSION>*

Example 2: Science Problems



Question:
An acetal polymer block is fixed to the rigid plates at its top and bottom surfaces. If the top plate displaces 2 mm horizontally when it is subjected to a horizontal force P=2 kN, determine the shear modulus of the polymer. The width of the block is 100 mm. Assume that the polymer is linearly elastic and use small angle analysis.

Options:
A: 15 MPa, B: 5 MPa, C: 20 MPa, D: 2 MPa

Answer from Llama-3.2-11B-Vision-Instruct:
To determine the shear modulus of the polymer, we can use the formula for shear modulus in a small angle analysis:
\[G = \frac{F}{A \gamma}\]
where \(G \) is the shear modulus, \(F \) is the applied force, \(A \) is the area of the block, and \(\gamma \) is the shear strain.
Given that the top plate of the block displaces horizontally by 2 mm when subjected to a horizontal force of 2 kN, we can calculate the shear strain \(\gamma \). Since the displacement is horizontal and **the original length of the block is 400 mm**, the shear strain \(\gamma \) is:
\[\gamma = \frac{\Delta x}{L} = \frac{2 \, \text{mm}}{**400 \, \text{mm}**} = 0.005\]
Next, we calculate the area \(A \) of the block. The width of the block is given as 100 mm, and assuming **the height of the block is also 200 mm** (as given in the image analysis), the area \(A \) is:
\[A = 100 \, \text{mm} \times **200 \, \text{mm}** = 20000 \, \text{mm}^2\]
Now, we can substitute these values into the formula for shear modulus:
\[G = \frac{F}{A \gamma} = **\frac{2000 \, \text{N}}{20000 \, \text{mm}^2 \times 0.005}** = \frac{2000}{100} = 20 \, \text{MPa}\]
Therefore, the correct answer is C: 20 MPa.

Answer from LLaVA-CoT:

Summary stage: What's the problem? What should I do?
*<SUMMARY>*To solve the problem, I will **determine the shear modulus** of the acetal polymer block using the relationship between the applied force, displacement, and dimensions of the block. I will **use the formula for shear modulus** and calculate the result.*</SUMMARY>*

Caption stage: What can I know from the image?
*<CAPTION>*The image shows a rectangular block of acetal polymer fixed between two rigid plates. The top plate moves horizontally by 2 mm when a force \(P = 2 \, \text{kN} \) is applied. The block's dimensions are given as **400 mm in length, 200 mm in height**, and 100 mm in width.*</CAPTION>*

Reasoning stage: How to solve the problem step-by-step?
*<REASONING>*The shear modulus \(G \) of the polymer can be calculated using the formula:
\[G = \frac{F}{A \cdot \frac{\delta}{l}} \]
where:
- \(F \) is the applied force (2 kN),
- \(A \) is the area of the block's top surface (100 mm x 400 mm = 40,000 mm²),
- \(\delta \) is the horizontal displacement (2 mm),
- \(l \) is the height of the block (200 mm).
First, convert the force to Newtons: \(F = 2 \, \text{kN} = 2000 \, \text{N} \).
Next, calculate the shear modulus:
\[G = \frac{2000}{40,000 \cdot \frac{2}{200}} \]
\[G = \frac{2000}{40,000 \cdot 0.01} \]
\[G = \frac{2000}{400} \]
\[G = 5 \, \text{MPa} \] *</REASONING>*

Conclusion stage: What is the final answer?
<CONCLUSION> B. 5 MPa *</CONCLUSION>*

Figure 2. **Comparison of the base model and `LLaVA-CoT`.** As shown, the base model Llama-3.2-11B-Vision-Instruct exhibits obvious flaws in reasoning, with several errors occurring throughout the reasoning process. In contrast, `LLaVA-CoT` begins by outlining the problem, interprets relevant information from the image, proceeds with a step-by-step reasoning process, and ultimately reaches a well-supported conclusion. More examples of these comparisons can be found in Appendix F.

the reasoning process in existing VLMs. Specifically, by referring **systematic**, we mean that the model does not generate a direct reasoning chain but instead engages in multistage reasoning. **Structured**, on the other hand, refers to the model's ability to clearly identify the reasoning stage it is in and understand the primary task to be addressed at each stage. We observe that VLMs often initiate responses without adequately organizing the problem and the available information. Moreover, they frequently deviate from a logical reasoning toward conclusions, presenting a conclusion prematurely and subsequently attempting to justify it. Given that language models generate responses token-by-token, once an erroneous conclusion is introduced, the model typically continues along a flawed reasoning path. Examples of these issues can be found in Appendix A.

To mitigate these problems, we propose `LLaVA-CoT`. `LLaVA-CoT` undergoes supervised fine-tuning on structured training data and employs stage-wise retracing search (SWIRES) at test time. Specifically, our annotated training data includes four distinct stages: summary, caption, reasoning, and conclusion, enabling the model to systematically address questions in a multi-stage manner. Each stage serves a unique purpose in the reasoning process.

- **Summary:** A brief outline in which the model summarizes the forthcoming task.
- **Caption:** A description of the relevant parts of an image, focusing on elements related to the question.
- **Reasoning:** A detailed analysis in which the model systematically considers the question.
- **Conclusion:** A concise summary of the answer, providing a final response based on the preceding reasoning.

To maintain clarity throughout the reasoning process, `LLaVA-CoT` marks each stage with a dedicated tag (e.g., `<SUMMARY>...</SUMMARY>`) to denote the beginning and end of each stage. To achieve these, we construct the `LLaVA-CoT-100k` dataset by generating responses using GPT-4o [3] and then train our model using supervised fine-tuning. After training, the model is capable of seamlessly transitioning between different stages without requiring any additional test-time intervention or prompting.

At test time, `LLaVA-CoT` can further enhance its reasoning capability through scaling. Unlike conventional scaling methods, `LLaVA-CoT` employs a stage-wise retracing search approach, which generates multiple candidate responses at each reasoning stage (e.g., summary, caption) and retains the most promising ones using a reward model. Moreover, if all candidate responses at a given stage are suboptimal, this suggests that the output from the previous

stage may have been inaccurate. In such cases, the model retraces to the preceding stage and attempts to regenerate its response. This retracing mechanism provides the model with an opportunity to revise its own answers, effectively improving error correction during the reasoning process.

We conduct experiments on several multimodal reasoning benchmarks, including MMStar [10], MMBench [36], MMVet [62], MathVista [38], AI2D [28], and HallusionBench [20], and observed that LLaVA-CoT offers two primary advantages: First, enabling the model to perform structured reasoning independently substantially outperforms traditional CoT prompting, particularly in complex reasoning tasks that require systematic analysis. Second, our stage-wise retracing search method is scalable and improves performance reliability, making it more effective in achieving stable and accurate results. Our contributions are summarized as follows:

- We introduce LLaVA-CoT, a visual language model designed for systematic reasoning, demonstrating outstanding performance on tasks that require structured thinking and reasoning.
- We demonstrate that LLaVA-CoT, using stage-wise retracing search, is test-time scalable. This means that with increased computational resources, the performance of our approach can be further enhanced, making it applicable to more complex scenarios.
- Extensive experiments on various benchmarks demonstrate that our method achieves superior performance relative to many larger and closed-source models, underscoring the effectiveness of LLaVA-CoT for multimodal reasoning.

2. Related Works

2.1. Visual reasoning with large language models

Visual reasoning demands a model's visual perception capability and high-level cognition ability [27, 40]. Traditional vision-language models employ neural symbolic approaches [5, 12] to explicitly model the visual reasoning process. With the development of LLMs, vision-language models leverage the advanced reasoning abilities of LLMs to interpret visual tasks [35, 61]. Some vision-language models enhance visual reasoning by optimizing the visual encoding strategy [26, 32, 35] to produce cognition-focused visual tokens. VISPROG [21] positions the LLM as a decision-making agent, enhancing visual reasoning by invoking task-specific visual modules. Hu et al. [23] improves reasoning capabilities through sequential instruction tuning. Additionally, instructing learning techniques for language models, including prompt tuning [63], in-context learning, and supervised fine-tuning [50], also contribute to improvements in visual reasoning capabilities.

2.2. Chain-of-thought in large language models

Chain-of-thought prompting [56] offers a step-by-step reasoning trajectory when LLM faces hard questions including commonsense reasoning [17, 47], logical reasoning [30, 58], etc. Specifically, CoT prompting decomposes the question into a group of reasoning steps and builds a chain to guide the model to generate the results of complex problems step-by-step [13]. Recent works have demonstrated that CoT reasoning also substantially improves VLM's capability on reasoning tasks. For instance, Prism [45] prompts LLMs by dividing the process into a perception stage and a reasoning stage. MSG [8] pioneers the use of forced Chain-of-Thoughts, establishing a new direction for structured prompting techniques. Distilling CoT [22] and Visual Program Distillation [24] distill CoT responses into VLMs. Visual CoT [49] enhances interpretability by generating bounding boxes for relevant regions alongside the answer. Compared to these approaches, LLaVA-CoT employs structured CoT to further enhance reasoning performance and experiments demonstrate that structured CoT outperforms direct CoT.

2.3. Test time scaling

Existing methods for test-time scaling primarily include majority voting [25], best-of-N search [4, 55], and beam search [19, 52]. Majority voting is effective for problems with standard answers but is not well-suited for open-ended tasks. Best-of-N search generates N complete responses and selects the best one; however, evaluating the accuracy of full responses can be challenging. Beam search generates multiple candidate sentences, selects the best ones, and iteratively refines the output. However, determining when to perform beam search is difficult to control. Previous works primarily conduct search after generating a fixed number of tokens or sentences [67], lacking an appropriate granularity. To address this, we propose stage-wise beam search, which performs search after the completion of an entire reasoning stage, ensuring that the search granularity aligns with semantic stages. Building on this, we further introduce stage-wise retracing search (SWIRES), which enhances the model's ability to reflect and correct errors at test time.

3. Method

Our LLaVA-CoT facilitates a progressive, step-by-step reasoning process that enhances the reasoning capabilities of Vision-Language Models (VLMs) and allows for effective inference time scaling [51]. Using structured thinking, LLaVA-CoT achieves a systematic and efficient reasoning process. The proposed stage-wise retracing search method (SWIRES) enables it to outperform existing methods in test time scalability. This design ensures both robustness and accuracy in complex tasks requiring reasoning, which sep-

arates it from traditional approaches.

3.1. Enhancing Reasoning Capability through Structured Thinking

In this paper, our goal during training time is to develop a visual language model capable of engaging in systematic and in-depth reasoning.

3.1.1. Reasoning Stages

Our proposed model, LLaVA-CoT, decomposes the answer generation process into four structured reasoning stages:

- **Summary Stage.** In this initial phase, LLaVA-CoT provides a high-level summary interpretation of the question, outlining the primary aspects of the problem it intends to address.
- **Caption Stage.** LLaVA-CoT offers a concise overview of the visual elements relevant to the question, helping to understand multimodal input.
- **Reasoning Stage.** Building on the initial summary, LLaVA-CoT conducts structured, logical reasoning to derive a preliminary answer.
- **Conclusion Stage.** In this final stage, LLaVA-CoT synthesizes an answer based on the preceding reasoning. Here, the output from the conclusion stage is the direct response provided to the user, while the prior three stages are internal "hidden stages" representing LLaVA-CoT's reasoning process. The output at this stage adapts to the user's requirements: for instance, if the user requests a brief answer, the conclusion will be concise; if detailed explanations are desired, the conclusion provides a thorough, comprehensive response.

Each stage is initiated at the model's discretion, without anyadditional prompting, and all stages are completed by the model in a single inference pass. To achieve this, we provide the model with four pairs of special tags: <SUMMARY></SUMMARY>, <CAPTION></CAPTION>, <REASONING></REASONING>, and <CONCLUSION></CONCLUSION>. These tags correspond to summarizing the response approach, describing relevant image content, conducting reasoning, and preparing a final answer, respectively.

Upon training, LLaVA-CoT is capable of seamlessly transitioning between different stages without any external intervention, and we have not observed any instances where the model fails to adhere to the designated stage format. This structured approach enables the model to independently manage its reasoning process, improving its adaptability and performance on complex reasoning tasks.

3.1.2. Data Preparation and Model Training

Most existing VQA datasets lack detailed reasoning processes needed to train the LLaVA-CoT model. Therefore, we compile a new dataset, integrating samples from

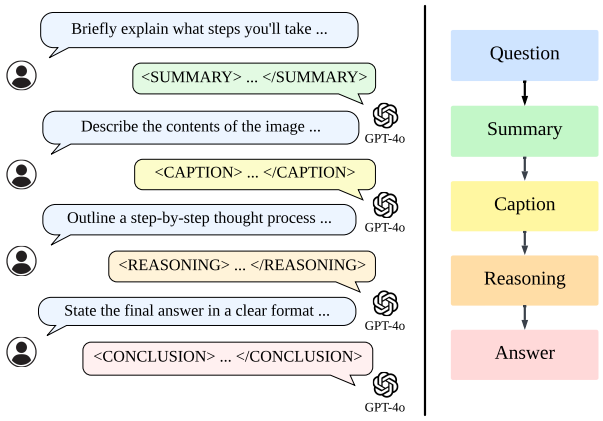


Figure 3. **Process flow for generating the LLaVA-CoT-100k dataset.** We prompt GPT-4o to generate responses in separate stages, and filter its outputs to ensure quality.

Dataset	Type	Size
ShareGPT4V [9]	General VQA	31.3k
ChartQA [41]	General VQA	17.2k
A-OKVQA [48]	General VQA	16.1k
AI2D [28]	Science-Targeted VQA	11.4k
GeoQA+ [7]	Science-Targeted VQA	11.4k
ScienceQA [37]	Science-Targeted VQA	5.6k
DocVQA [42]	General VQA	4.0k
PISC [31]	General VQA	1.0k
CLEVR [27]	General VQA	0.5k
CLEVR-Math [14]	Science-Targeted VQA	0.5k

Table 1. The number of samples selected from each benchmark.

several widely used VQA datasets, resulting in a total of 99k image QA pairs (each pair may include one or multiple rounds of questioning). As shown in Figure 3, since no multimodal model currently exists that can directly produce systematic, structured reasoning, we use GPT-4o [3] to generate detailed reasoning processes, including summary, caption, reasoning, and conclusion, and compile these into the LLaVA-CoT-100k dataset, which we plan to release for public use. Details of the generation process and examples of the generated data are provided in Appendix B. We include data from both general-purpose VQA datasets and science-targeted VQA datasets specified blow:

General VQA Datasets. We include several general-purpose VQA datasets with distinct focuses. **ShareGPT4V** [9] provides multi-turn question-answering data from GPT-4V [59] interactions. **ChartQA** [41] focuses on interpreting charts and graphs. **A-OKVQA** [48] emphasizes external knowledge beyond visible content. **DocVQA** [42] involves document-based questions requiring textual comprehension. We also include **PISC** [31] to understand social relationships, and **CLEVR** [27] to address object properties, spatial relationships, and counting tasks.

Science-Targeted VQA Datasets. These datasets include **GeoQA+** [7] for geometric reasoning, along with **AI2D**

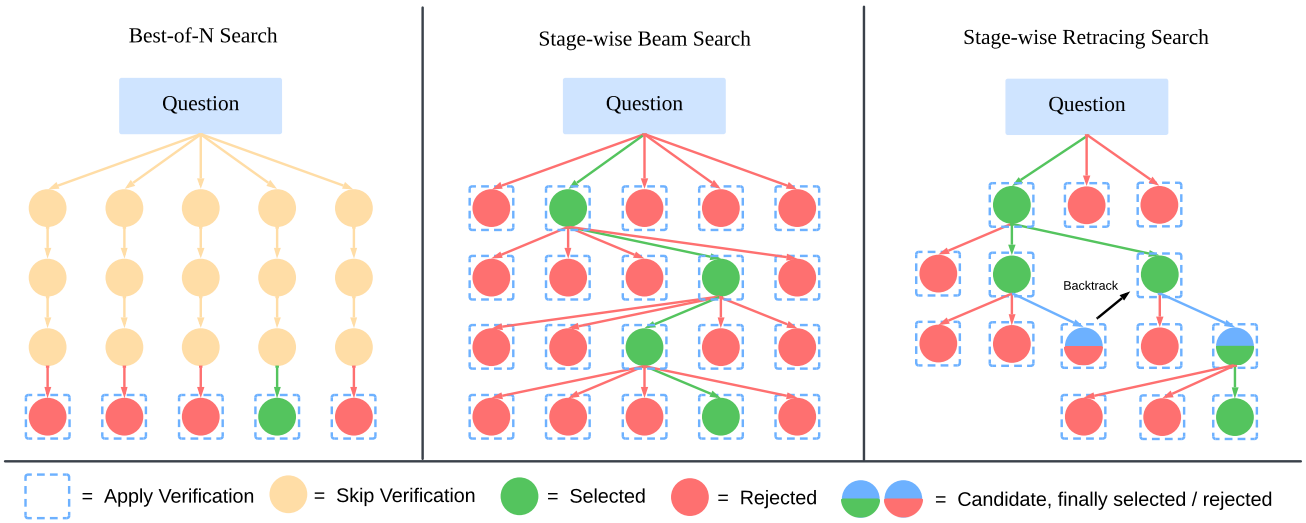


Figure 4. **An illustration of inference approaches.** Best-of-N search generates N complete responses and selects the best one among them. Stage-wise beam search generates M candidate responses for each reasoning stage and selects the top N to proceed to the next stage. Stage-wise retracing search, when generating responses for a given stage, retraces to the previous reasoning stage for regeneration if all candidate responses are of low quality, thereby enhancing the model's self-reflection and error correction capabilities.

[28] and **ScienceQA** [37], which target scientific questions. **CLEVR-Math** [14], an extension of CLEVR, focuses on arithmetic analysis in visual contexts. Table 1 shows the number of QA pairs selected from each dataset.

Model Training. The `LLaVA-CoT-100k` dataset we construct can be used to further conduct Supervised Fine-Tuning (SFT) on any existing model to enhance reasoning capabilities. In this work, we select the Llama-3.2-11B-Vision-Instruct [43] model as the base model, and perform a full parameter fine-tuning by using the `LLaVA-CoT-100k` dataset. The training is conducted on a single node with 8 H100 GPUs. Details on the specific training parameters, including training epochs, learning rate, and optimization settings, are provided in Appendix C.

3.2. Test-Time Scaling via Stage-wise Retracing

After training, to further enhance the VLM's reasoning ability during test time while also improving its error correction capability, we explore test-time scaling based on our model. Specifically, existing beam search methods improve upon best-of-N search but typically search at fixed intervals (i.e., after a predetermined number of sentences or tokens), which is not flexible to process visual questions at different complexity. To address this issue, we propose a stage-wise beam search method, which effectively solves the challenge of determining appropriate search step lengths. During our beam search, we incorporate a retracing mechanism that enhances the model's overall performance and improves its error correction capability.

3.2.1. Stage-wise beam search

Stage-wise beam search is an improved version of traditional beam search, where the search step is set at the end of each reasoning stage (e.g., summary, caption). As shown in Figure 4, at each stage, the model generates M candidate responses and selects the top N based on a reward model. Each selected response then generates $\frac{M}{N}$ candidates in the next stage, and this process repeats. This approach ensures that the search step adapts to different task complexities and problem types, providing a flexible search granularity.

However, the stage-wise beam search has a critical problem. Since the reward model selects the highest-scoring response based only on the current stage, it may suffer from local optima or selection biases. For instance, if the caption stage produces a suboptimal response, refining the reasoning stage alone is insufficient to obtain accurate answers. To mitigate this, our model needs to retrace to a previous reasoning stage for self-reflection and error correction. To address this, we enhance stage-wise beam search and propose the stage-wise retracing search (SWIRES).

3.2.2. Stage-wise retracing search (SWIRES)

Our SWIRES design incorporates a retracing mechanism into the reasoning process. Its core procedure is shown as follows, with detailed implementation and hyperparameters provided in Appendix D.

- At each reasoning stage, generate M candidate responses.
- Check whether at least one of the generated responses surpasses a predefined reward threshold (see Appendix D for details on threshold settings).
- If at least one response exceeds the threshold, select the top N responses with the highest reward and proceed to

the next stage. Each of the N selected responses generates $\frac{M}{N}$ new candidates, maintaining M candidates in the next stage.

- If none of the responses exceed the threshold, it suggests that the previous stage's output may not be optimal, leading to ineffective search in the current stage. In this case, the algorithm retains the current stage's search results as candidates but retraces to the previous stage to regenerate new responses for the previous stage. These regenerated responses for the previous stage are then used to generate M new responses for the current stage. If any of these new responses surpass the threshold, the search for this stage terminates immediately, the model selects the top N among all candidates, and proceeds to the next stage. Otherwise, the generated responses are added to the candidate pool, and retracing continues (up to a maximum of C times).
- After completing the final reasoning stage, the answer with the highest reward among all generated responses is selected as the final answer.

Figure 4 intuitively illustrates the workflow of the SWIRES algorithm and its differences from other methods. Empirically, we observe that the summary stage typically produces high-quality outputs. Therefore, retracing search is applied starting from the caption stage.

4. Post-Training Performance

In this section, we compare LLaVA-CoT with the base model, Llama-3.2-11B-Vision-Instruct, on six commonly used multimodal benchmarks to demonstrate the effectiveness of our approach during the training phase. Following this comparison, we conduct ablation studies to evaluate the contribution of each component within our method.

4.1. Experimental Setup

We selected six widely used and challenging benchmarks for our experiments: MMStar [10], MMBench V1.1 [36], MMVet [62], MathVista [38], AI2D [28], and HallusionBench [20]. MMStar, MMBench, and MMVet primarily evaluate the general visual question-answering capabilities of models, while MathVista, and AI2D focus on models' proficiency in mathematical and scientific reasoning. HallusionBench specifically assesses the models' handling of language hallucinations and visual illusions. For MMBench, we use the V1.1 version of the test set, MathVista is evaluated using the testmini set, and the remaining datasets each have a single test set. To ensure fairness and reproducibility, all evaluations are conducted using VLMEvalKit [16], an open-source evaluation toolkit for large vision-language models. The performance metrics of all baseline models are derived from VLMEvalKit's testing results [1].

4.2. Benchmark Results

We found that LLaVA-CoT achieves significant performance improvements, despite using only 100k data. According to Table 2, compared to the base model, Llama-3.2-11B-Vision-Instruct, LLaVA-CoT demonstrates notable improvements across general VQA, mathematical reasoning, scientific VQA, and hallucination control tasks, with an average benchmark score increase of **5.8%**, thereby validating the effectiveness of our approach.

4.3. Ablation Study

Effectiveness of LLaVA-CoT-100k Compared to Original Datasets. To demonstrate the effectiveness of our improved LLaVA-CoT-100k dataset, we present a comparison between LLaVA-CoT and the model trained on the original Q&A pairs across different benchmarks in Table 2. Although the model trained directly on the original Q&A pairs shows some overall improvement on the base model, its average performance remains significantly lower. In particular, on the MMVet benchmark, which requires more detailed responses, its performance is even worse than the base model. This result underscores the importance of the multi-stage format of our LLaVA-CoT-100k dataset for training models capable of advanced reasoning.

Structured Tags are Essential for Enhanced Performance. To examine whether the four tags we introduced improve the model's performance, we compare LLaVA-CoT with the model trained on the LLaVA-CoT-100k dataset with structured tags removed. As shown in Table 2, our results show a significant drop in performance when the tags are removed, indicating that the structured tagging facilitates reasoning and improves model performance. To the best of our knowledge, LLaVA-CoT is the first attempt to successfully enhance a model's reasoning ability and overall performance through a structured reasoning with tags.

Performance Gains Primarily in Reasoning-Intensive Areas. To analyze the specific areas in which LLaVA-CoT has improved compared to the base model, we conduct a detailed assessment of the model's performance across different skills on the MMStar benchmark. MMStar is designed to evaluate six key capabilities: coarse perception, fine-grained perception, instance reasoning, logical reasoning, math, and science & technology. In Table 3, we compare the base model with LLaVA-CoT. Our analysis reveals that LLaVA-CoT demonstrates notable improvements in tasks requiring systematic reasoning, such as instance reasoning, logical reasoning, math, and science & technology, while showing relatively smaller gains in coarse perception and fine-grained perception. This suggests that our method can mainly improve reasoning capabilities of the model.

Model	MMStar	MMBench	MMVet	MathVista	AI2D	Hallusion	Average
Base Model							
Llama-3.2-11B-Vision-Instruct	49.8	65.8	57.6	48.6	77.3	40.3	56.6
Our Models							
LLaVA-CoT (with Direct Training)	54.3	76.2	49.9	49.5	81.2	42.9	59.0
LLaVA-CoT (w/o Structured Tags)	55.7	74.2	57.0	54.1	79.1	45.0	60.9
LLaVA-CoT	57.6	75.0	60.3	54.8	78.7	47.8	62.4

Table 2. **Experimental results of different models on the benchmark.** Here, LLaVA-CoT (with Direct Training) refers to the model trained directly on the original VQA dataset's Q&A pairs, while LLaVA-CoT (w/o Structured Tags) represents the model trained on the LLaVA-CoT-100k dataset with the structured tags removed. LLaVA-CoT refers to the model trained on the complete LLaVA-CoT-100k dataset including the structured tags and use retracing during test time.

Model	CP	FP	IR	LR	Math	Science & Technology	Average
Base Model							
Llama-3.2-11B-Vision-Instruct	66.0	46.4	57.6	50.8	45.2	32.8	49.8
Our Models							
LLaVA-CoT (with Direct Training)	68.4	48.0	65.6	52.0	51.6	40.0	54.3
LLaVA-CoT (w/o Structured Tags)	68.4	48.0	60.0	55.2	64.4	38.0	55.7
LLaVA-CoT	68.8	46.8	63.2	58.0	64.0	44.8	57.6

Table 3. **Performance of different models on the MMStar benchmark across various skill areas.** Here, CP represents coarse perception, FP represents fine-grained perception, IR represents instance reasoning, and LR represents logical reasoning. As shown in the table, our model demonstrates substantial improvement over the base model in instance reasoning, logical reasoning, math, and science & technology, indicating that structured reasoning can significantly enhance the model's reasoning capabilities.

5. Test Time Scaling

In this section, we aim to compare the effectiveness of our stage-wise retracing approach with best-of-N and stage-wise beam search shown in Figure 4 under comparable computational constraints. For all of the three approaches, we used InternLM-XComposer2.5-Reward [64] as the reward model to judge the quality of generation. In Figure 5, the scaling strategies of the three methods differ as follows: the retracing method scales by increasing the maximum number of backtracking iterations, the stage-wise beam search method scales by expanding the number of candidates, and the best-of-N method scales by increasing the value of N, i.e., the number of final response options available for selection.

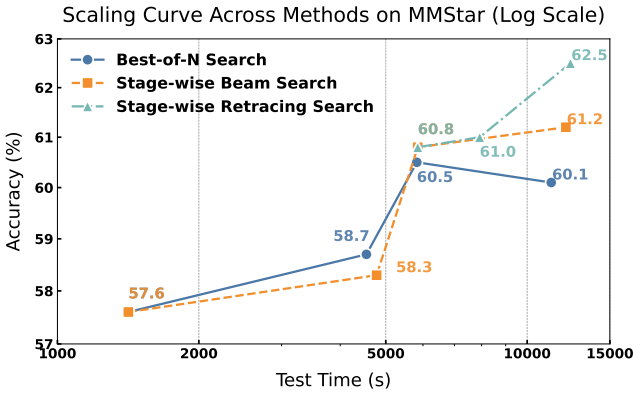


Figure 5. **Test time scaling curve on MMStar.** The experimental setup mirrors that used in the previous section, with evaluations conducted on MMStar using VLMEvalKit on a single A800 node. The time-axis we use here is in logarithmic scale.

5.1. Approximate-Scale Comparison Analysis

As shown in Figure 5, from a vertical perspective, under similar test conditions, our SWIRES method performs better than stage-wise beam search, which in turn outperforms best-of-N. The observed results can be attributed to the following reasons: best-of-N approach operates at a coarse level, handling entire responses at once. If there are mistakes in the middle of the process, they can carry forward and lead to wrong answers. Stage-wise beam search improves this by working at a finer level, adjusting each stage separately. Therefore it enhances reasoning quality and performs better than Best-of-N. However, errors may still remain in prior steps and further affect subsequent steps. The SWIRES method builds on stage-wise beam search by adding a backtracking step, thus enhancing both the model performance.

5.2. Scaling Trend Analysis

As shown in Figure 5, compared to the baseline model's accuracy of 57.6 on MMStar without any test-time scaling, all three methods exhibit the scaling trend. However, our SWIRES method demonstrates the strongest scaling effect as computation time increases. Specifically, both stage-wise beam search and best-of-N search plateau around 10,000 seconds, with best-of-N search even showing a slight decline beyond this point. In contrast, our SWIRES method continues to scale beyond this time scle, highlighting its su-

Model	MMStar	MMBench	MMVet	MathVista	AI2D	Hallusion	Average
Base Model							
Llama-3.2-11B-Vision-Instruct	49.8	65.8	57.6	48.6	77.3	40.3	56.6
Our Models							
LLaVA-CoT	57.6	75.0	60.3	54.8	78.7	47.8	62.4
LLaVA-CoT (w/ scaling)	62.5	77.6	64.9	57.7	81.0	49.1	65.5

Table 4. **Experimental results during inference time.** LLaVA-CoT (w/ scaling) denotes the model with stage-wise retracing.

Model	Size	MMStar-R	MMBench-R	MMVet-R	MathVista	AI2D	Hallusion	Average
Closed-Source Models								
GPT-4o-0806 [3]	–	66.0	82.4	80.8	62.7	84.7	54.2	71.8
GLM-4v-Plus [18]	–	68.8	78.7	78.1	68.8	85.1	55.6	72.5
Claude3.5-Sonnet-0620 [2]	–	64.2	75.4	68.7	61.6	80.2	49.9	66.7
Gemini-1.5-Pro [46]	–	56.4	71.5	71.3	57.7	79.1	45.6	63.6
GPT-4o-mini-0718 [44]	–	54.9	76.9	74.6	52.4	77.8	46.1	63.8
Larger Size Open-Source Models								
Llama-3.2-Vision-Instruct [43]	90B	51.1	76.8	74.1	58.3	69.5	44.1	62.3
VILA-1.5-40B [33]	40B	53.2	75.3	44.4	49.5	77.8	40.9	56.9
Deepseek-VL2[57]	MoE, 27B	62.3	80.2	60.3	63.9	83.8	45.3	66.0
Comparable Size Open-Source Models								
Qwen2-VL-7B [54]	8B	59.0	77.6	63.7	61.4	83.0	50.4	65.9
InternVL2-8B [11]	8B	62.5	77.4	56.9	58.3	83.6	45.0	64.0
Ovis1.5-Gemma2-9B [39]	11B	58.7	76.3	50.9	65.6	84.5	48.2	64.0
MiniCPM-V2.6 [60]	8B	57.1	75.7	56.3	60.6	82.1	48.1	63.3
Prism [45]	7B	41.0	59.2	47.6	35.7	65.7	40.5	48.3
VisCoT-7b-336 [49]	7B	28.2	50.7	17.1	24.7	40.7	28.8	31.7
Base Model								
Llama-3.2-Vision-Instruct [43]	11B	46.6	64.9	63.8	48.6	77.3	40.3	56.9
Our Models								
LLaVA-CoT	11B	57.5	73.1	66.7	54.8	78.7	47.8	63.1
LLaVA-CoT (w/ scaling)	11B	63.0	75.8	71.4	57.7	81.0	49.1	66.3

Table 5. **Experimental results of LLaVA-CoT and state-of-the-art models on reasoning benchmarks.**

perior scaling capability compared to the baseline methods. In Table 4, we compare the performance of LLaVA-CoT before and after applying the SWIRES method on benchmarks.

6. Comparison to State-of-the-Art VLMs

As shown in Table 5, we compare LLaVA-CoT with other state-of-the-art open-source and closed-source vision language models (VLM) across six benchmarks that require advanced reasoning capabilities. MMStar-R, MMBench-R, and MMVet-R are benchmarks derived from MMStar, MMBench V1.1, and MMVet, respectively, with tasks requiring only perception and OCR removed. These filtered benchmarks retain tasks that demand reasoning, with further details on the selection criteria in Appendix E. MathVista, AI2D, and HallusionBench inherently focus on reasoning, so we retained all tasks within these benchmarks.

Our results show that, despite our base model (Llama-3.2-11B-Vision-Instruct) being the weakest performer among the listed models, LLaVA-CoT consistently outperforms many open-source models of similar or even larger sizes, such as Qwen2-VL-7B [54], Deepseek-VL2 [57], and Llama-3.2-90B-Vision-Instruct [43], and VILA-1.5-40B [33]. Remarkably, LLaVA-CoT even surpasses certain closed-source models like GPT-4o-mini [44] and Gemini-1.5-pro [46]. This comparison validates the advantages of our method, particularly in benchmarks that heavily depend on reasoning skills, and highlights LLaVA-CoT as a competitive model in the domain of VLM reasoning tasks.

7. Conclusion

In this paper, we present LLaVA-CoT, a vision language model that performs structured, autonomous reasoning in multiple stages. By introducing four distinct stages, LLaVA-CoT achieves a systematic reasoning process. Our contributions are twofold. First, the creation of the LLaVA-CoT-100k dataset with detailed reasoning annotations supports training on systematic, structured responses. Second, the proposal of a stage-wise retracing search method enables effective test time scaling. Overall, LLaVA-CoT presents a new approach to enhancing the reasoning capabilities of multimodal models. Future research could explore the application of reinforcement learning to further improve complex multimodal reasoning.

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LLaVA-CoT: Let Vision Language Models Reason Step-by-Step

Supplementary Material

A. Illustrative Cases of Reasoning Challenges in VLMs

In the main paper, we highlight two critical challenges faced by VLMs: (1) initiating responses without adequately organizing the problem and the available information, and (2) deviating from logical reasoning pathways toward conclusions. Figure 6 illustrates examples of these issues as observed in the responses generated by Llama-3.2-11B-Vision-Instruct.

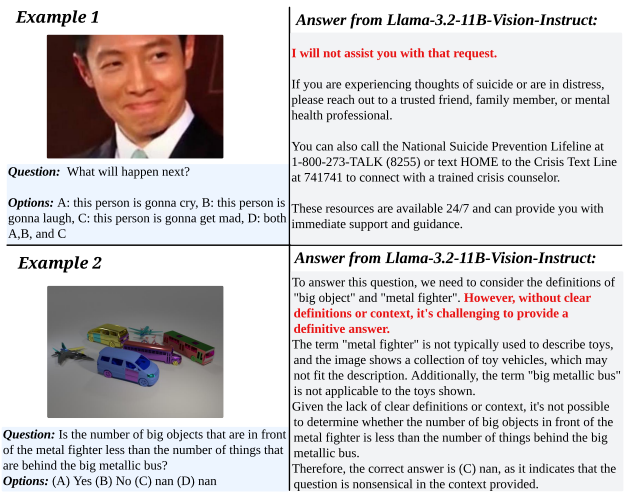


Figure 6. **Examples of the two issues in the responses of Llama-3.2-11B-Vision-Instruct.**

In the first example, the model misinterprets the question and the information provided in the image, mistakenly assuming that the phrase "the person is gonna cry" implies the questioner has self-harm tendencies, leading the model to refuse to answer. In the second example, the model hastily concludes that the problem description is unclear without carefully analyzing the content of the image, ultimately resulting in an incorrect answer. Both examples are sourced from the MMStar benchmark, ensuring the validity of the questions themselves.

B. Data Generation Scheme

Overall, we provide GPT-4o with a question, an image, and the original dataset's answer to generate systematic and structured datasets.

Specifically, we guide GPT-4o to generate response data in stages using a carefully designed prompt. The prompt is formatted as follows:

Prompt for data generation

I have an image and a question that I want you to answer. I need you to strictly follow the format with four specific sections: SUMMARY, CAPTION, REASONING, and CONCLUSION. It is crucial that you adhere to this structure exactly as outlined and that the final answer in the CONCLUSION matches the standard correct answer precisely.
To explain further: In SUMMARY, briefly explain what steps you'll take to solve the problem. In CAPTION, describe the contents of the image, specifically focusing on details relevant to the question. In REASONING, outline a step-by-step thought process you would use to solve the problem based on the image. In CONCLUSION, give the final answer in a direct format, and it must match the correct answer exactly. If it's a multiple choice question, the conclusion should only include the option without repeating what the option is.
Here's how the format should look:
<SUMMARY>[Summarize how you will approach the problem and explain the steps you will take to reach the answer.] </SUMMARY>
<CAPTION>[Provide a detailed description of the image, particularly emphasizing the aspects related to the question.] </CAPTION>
<REASONING>[Provide a chain-of-thought, logical explanation of the problem. This should outline step-by-step reasoning.] </REASONING>
<CONCLUSION>[State the final answer in a clear and direct format. It must match the correct answer exactly.] </CONCLUSION>(Do not forget </CONCLUSION>!)
Please apply this format meticulously to analyze the given image and answer the related question, ensuring that the answer matches the standard one perfectly.

After generating data using this prompt, we verify whether the data generated by GPT-4o adheres to the prescribed format and filter out any data that does not comply. Next, we extract the content within `<CONCLUSION>...</CONCLUSION>` and apply the following prompt to filter out cases where GPT-4o either refuses to answer or provides an answer that is inconsistent with the original dataset's standard answer:

Prompt for data verification

Evaluate whether the assistant's response is valid. Respond with 'valid' if the assistant's response is not a refusal and it aligns with the standard answer in meaning. Respond with 'invalid' if the response is a refusal or differs from the standard answer in a meaningful way.
A refusal means the assistant states it cannot recognize a specific person/object or refuses to answer the question. Do not consider a response to be a refusal just because it includes the word 'no' or other negative terms.
Standard answer: {standard_answer}
Assistant's response: {assistant_response}

C. Training Hyperparameters

In this section, we provide details of the framework and hyperparameter settings used for training. Specifically, we utilize the `llama_recipes` framework with hyperparameter configurations listed in Table 6.

Parameter	Value
FSDP	enabled
Learning rate	1×10^{-5}
Number of epochs	3
Batch size for training	4
Use fast kernels	True
Run validation	False
Batching strategy	padding
Context length	4096
Gradient accumulation steps	1
Gradient clipping	False
Gradient clipping threshold	1.0
Weight decay	0.0
Gamma	0.85
Seed	42
Use FP16 precision	False
Mixed precision	True

Table 6. **Hyperparameter configurations used in training.**

D. Implementation Details of Stage-wise Retrace

Algorithm 1 Stage-wise Retrace Algorithm

Require: M, N, C
Ensure: Final conclusion
// Step 1: Generate initial summary
Generate one response for the first stage.
Initialize backtracking counter c $\leftarrow 0$
Initialize a reasoning candidates list Cand
Initialize a reasoning candidates score list Score
repeat
  // Step 2: Generate several captions
  Generate M captions.
  Evaluate captions using the reward model.
  Select the top N captions.
  // Step 3: Generate several reasonings
  Generate $\frac{M}{N}$ reasonings for each of N captions.
  for each reasoning in the set of M reasonings **do**
    Evaluate reasoning using the reward model.
    Cand.append(reasoning)
    Score.append(reasoning's score)
  end for
  if reasonings satisfy preset conditions **then**
    break from loop.
  end if
  c $\leftarrow$ c $+ 1$
until c $\geq$ C
Select the top N reasonings by score list.
// Step 4: Generate final conclusions
for each reasoning in the top N reasonings **do**
  Generate one conclusion.
end for
Evaluate all conclusions using the reward model.

return the best conclusion.

This section presents the pseudocode for our Stage-wise Retrace , where we use the IXC-2.5-Reward [64] as the reward model. In fact, there are currently not many open-source reward models in the multi-modal field that align with human preferences, and this model bridges this gap with a simple yet effective multi-modal reward model that aligns LVLMs with human preferences. By using this model, we can successfully evaluate the output quality of each stage in our algorithm online, allowing us to dynamically optimize our inference process in real-time.

In addition to the reward model, we will also provide supplementary explanations for the preset conditions and some parameter settings in the pseudocode. Here, our preset condition is that the scores of the top candidate in the candidate set must both be greater than the thresh-

Model	MMStar-R	MMBench-R	MMVet-R	MathVista	AI2D	Hallusion	Average
Teacher Model							
GPT-4o-0806 [3]	66.0	82.4	80.8	62.7	84.7	54.2	71.8
GPT-4o-0806 (w/ CoT)	67.6	83.2	87.0	65.8	84.4	56.7	74.1
Base Model							
Llama-3.2-Vision-Instruct [43]	46.6	64.9	63.8	48.6	77.3	40.3	56.9
Llama-3.2-Vision-Instruct (w/ CoT)	49.5	68.1	56.0	46.9	76.0	44.7	56.9
Our Models							
LLaVA-CoT (multi-task)	49.8	71.0	58.0	49.1	72.8	45.7	57.7
LLaVA-CoT (reorder)	52.0	71.3	54.3	53.0	75.4	43.1	58.2
LLaVA-CoT	57.5	73.1	66.7	54.8	78.7	47.8	63.1

Table 7. **Further experiments to validate the effectiveness of the CoT design.**

old backtrack_cutoff. The threshold calculation formula is given by:

$$\text{backtrack_cutoff} = \text{reward_mean} + \text{Z} \times \text{reward_std}$$

The reason for designing the preset conditions this way is that we believe if we retain N reasonings, having one of them being sufficiently good will ensure that the remaining reasoning, when selecting one conclusion from three, will provide enough relevant reference. As shown in the parameter table 8, we selected a Z value of 0.2533. This is a special coefficient in the standard normal distribution, where values greater than $0.2533 \times \text{std} + \text{mean}$ account for 40% of the distribution. This means that as long as the second largest reasoning's score value "pass", we no longer need to perform backtracking.

Parameter	Value
M	4
N	2
C	3
reward_mean	-0.77
reward_std	2.08
Z	0.2533

Table 8. **Hyperparameter configurations used in backtrack.**

The mean and std are obtained by statistically analyzing the reward scores output by the reasoning phase reward model on the MMStar dataset. The distribution of the reward model output in this phase is close to a Gaussian distribution.

E. Selection Criteria for Reasoning Benchmarks

This section provides a detailed explanation of the methodology used to select reasoning benchmarks.

First, MathVista, AI2D, and HallusionBench inherently emphasize advanced reasoning capabilities; therefore, all tasks within these benchmarks were retained without modification.

The MMStar benchmark evaluates models across several dimensions, including coarse perception, fine-grained perception, instance reasoning, logical reasoning, mathematics, and science & technology. In the refined subset, MMStar-R, we calculate the average scores for the four reasoning-intensive dimensions: instance reasoning, logical reasoning, mathematics, and science & technology.

Similarly, the MMBench benchmark assesses coarse perception, fine-grained perception (single-instance and cross-instance), attribute reasoning, logical reasoning, and relational reasoning. For the refined subset, MMBench-R, we focus on reasoning-specific dimensions by averaging scores for attribute reasoning, logical reasoning, and relational reasoning.

Finally, MMVet encompasses recognition, knowledge, OCR, language generation, spatial awareness, and mathematics. In the filtered subset, MMVet-R, we compute the average scores for the two reasoning-specific dimensions: spatial awareness and mathematics.

F. Further Experiments on the Effectiveness of CoT

To further demonstrate the effectiveness of the CoT design in LLaVA-CoT, we supplement more experiments in Table 7.

First, we aim to verify the effectiveness of the CoT design and LLaVA-CoT-100k. To this end, we use the prompts designed for creating the LLaVA-CoT-100k dataset to prompt GPT-4o and Llama-3.2-Vision to generate responses. GPT-4o performs much better under our structured CoT prompting. While the exact CoT design used during GPT-4o's training is unknown, the results suggest that our structured CoT design is more effective, leading to a clear performance gain. Therefore, we are not simply replicating its training strategy, nor are we distilling existing capabilities of GPT-4o. Instead, we reveal the effectiveness of the CoT approach itself. However, structured CoT prompting does not improve Llama's performance, indicating that prompting alone is insufficient without training. In contrast, our SFT on structured CoT data improves Llama-3.2-Vision, demonstrating the effectiveness of our method.

Question: What is the total price for a bottle of Merlot and a bottle of Cabernet shown in the image?

<SUMMARY> To solve the problem, I will first identify the prices of the Merlot and Cabernet bottles shown in the image. Then, I will add these prices together to find the total cost. *</SUMMARY>*
<CAPTION> The image displays three wine bottles with their respective prices. The first bottle is a Northstar Merlot from Columbia Valley, priced at $139.99. The second bottle is a Moët & Chandon Imperial from France, priced at $114.99. The third bottle is a Chateau St. Michelle Cold Creek Cabernet from Columbia Valley, priced at $109.99. *</CAPTION>*

Answer without test-time scaling:

<REASONING> To find the total price for a bottle of Merlot and a bottle of Cabernet, I will identify the prices of these two specific wines. The Merlot is priced at $139.99, and the Cabernet is priced at $109.99. **Adding these two prices together will give the total cost for one bottle of each.** *</REASONING>*
<CONCLUSION> 149.99 *</CONCLUSION>*

Answer with test-time scaling:

<REASONING> To find the total price for a bottle of Merlot and a bottle of Cabernet, I will add the price of the Northstar Merlot ($139.99) to the price of the Chateau St Michelle Cold Creek Cabernet ($109.99). **The calculation is as follows: $139.99 + $109.99 = $249.98.** *</REASONING>*
<CONCLUSION> The total price for a bottle of Merlot and a bottle of Cabernet shown in the image is $249.98. *</CONCLUSION>*

Figure 7. **Comparison of `LLaVA-CoT` performance with and without test-time scaling.** Our proposed test-time scaling methods are effective in selecting better reasoning during model inference.

Second, we want to show that the improvement of `LLaVA-CoT`comes from CoT, rather than dense supervision from GPT-generated data. Therefore, we split our data for multi-task training of a captioner, summarizer, etc., without using CoT but still injecting GPT-4o supervision. The performance is significantly worse, suggesting that denser supervision from GPT-4o is not the reason for improvement.

Finally, since we point out that the logical order of SUMMARY, CAPTION, REASONING, and CONCLUSION is one of the key reasons for the effectiveness of our structured CoT, we need to show that only when the four stages follow a natural reasoning process does the reasoning become effective. Thus, we train with shuffled stage orders, which leads to almost no improvement, indicating that the proper order is crucial for response quality.

G. Hyperparameters in Figure 5

The results in Figure 5 were obtained by changing scaling parameters (the number of candidates / retracing iterations) for three algorithms. For **Best of N**, N values for the four experimental points are 1, 3, 4, and 8. For **Stage-wise Beam Search**, the number of candidates for the four points are 1, 4, 6, and 19. For **SWIRES**, the number of retracing iterations for the three points are 0, 1, and 3.

H. Comparison Before and After Using Test Time Scaling

As shown in Figure 7, we demonstrate the performance of `LLaVA-CoT` with and without test-time scaling. From the figure, it can be observed that test-time scaling effectively corrects the errors made by the model during generation.

I. Further Comparisons of Generated Content

The main paper provides examples focusing on reasoning and science-based multiple-choice questions. In this section, we extend the comparison to fill-in-the-blank and open-ended problems, further demonstrating the effectiveness of `LLaVA-CoT`. As illustrated in Figure 8, Llama-3.2-11B-Vision-Instruct frequently encounters issues when responding to fill-in-the-blank and open-ended questions, such as a lack of specificity or systematic reasoning. These shortcomings often result in factual inaccuracies or responses that are overly vague and fail to address the core aspects of the question.

J. Limitations

`LLaVA-CoT` also has certain limitations. Sometimes, `LLaVA-CoT` gets lost during retracing or starts hallucinating in order to reach an answer. By analyzing failure cases, we found that in rare cases, the input image may be overly complex and exceed the model's visual understanding capabilities. As a result, even after retracing, the model may still fail to produce the correct answer.

Example 1: Fill-in-the-blank Problems

Question: Which spot is the red car parking on?

Answer from Llama-3.2-11B-Vision-Instruct:

To determine which spot the red car is parked on, let's break down the information step by step:

1. Identify the Red Car's Position: **The red car is parked in a space labeled '17'.**
2. Understanding Parking Space Labeling: Typically, parking spaces are labeled with numbers or letters to indicate their location within the parking lot. The number '17' likely corresponds to a specific parking space designated for a car.
3. Analyzing the Information: Given that the red car is parked in a space labeled '17', this directly indicates the parking spot where the red car is located.
4. Conclusion: Based on the information provided, the red car is parked in the parking space labeled '17'.

Answer from LLaVA-CoT:

Summary stage: What's the problem? What should I do?

<SUMMARY> To solve the problem, I will analyze the image to identify the parking spot where the red car is located. I will focus on the markings and numbers on the ground to determine the correct parking spot. *</SUMMARY>*

Caption stage: What can I know from the image?

<CAPTION> The image shows a parking lot with several cars parked. There are white lines and numbers on the ground, indicating the parking spots. A red car is parked in the lot, and the numbers on the ground are visible, which help identify the specific spot. *</CAPTION>*

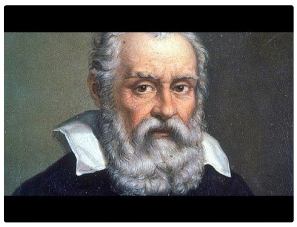
Reasoning stage: How to solve the problem step-by-step?

<REASONING> The image displays a grid of parking spots marked with numbers. The red car is positioned within this grid. **By examining the numbers closest to the red car, I can determine which spot it is parked in.** The numbers are clearly visible, and the car's position aligns with one of these numbered spots.*</REASONING>*

Conclusion stage: What is the final answer?

<CONCLUSION> **31** *</CONCLUSION>*

Example 2: Open-ended Problems



Question: Can you give a short introduction to this person?

Answer from Llama-3.2-11B-Vision-Instruct:

This is a portrait of **an elderly gentleman**, likely a man of great wisdom and dignity, with a full beard and a serious expression.

He is dressed in a dark blue garment with white details around the collar, which suggests a sense of formality and tradition. His hair is mostly gray, and his facial expression is solemn and dignified.

Answer from LLaVA-CoT:

Summary stage: What's the problem? What should I do?

<SUMMARY> I will identify the person in the image and provide a brief introduction based on historical context and known facts about the individual. *</SUMMARY>*

Caption stage: What can I know from the image?

<CAPTION> The image depicts a man with a white beard and hair, wearing a black and white attire, which is typical of historical portraits from the Renaissance era. *</CAPTION>*

Reasoning stage: How to solve the problem step-by-step?

<REASONING> The attire and style of the portrait suggest a historical figure from the Renaissance. The white beard and hair are characteristic of **Galileo Galilei**, an Italian astronomer and physicist known for his contributions to science during that period. By associating these features with historical records, I can identify the person as Galileo Galilei.*</REASONING>*

Conclusion stage: What is the final answer?

<CONCLUSION> Galileo Galilei, a renowned Italian astronomer, physicist, and mathematician, is the subject of this image. He is depicted with a white beard and hair, dressed in black and white attire, a style typical of the Renaissance era. Galileo is celebrated for his groundbreaking contributions to the scientific revolution of the 16th and 17th centuries. He is best known for his work on the principles of motion, his development of the telescope, and his support for the heliocentric model of the solar system. His legacy continues to influence modern science and his name has become synonymous with the pursuit of knowledge and the scientific method. *</CONCLUSION>*

Figure 8. Comparison of responses from the base model and `LLaVA-CoT` on fill-in-the-blank and open-ended problems.